Vegetation and Wildfire Risk Changes in Butte County, 2011 - 2021

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Introduction

Wildfires have become more frequent and destructive in California in recent decades, likely due to changing climate conditions. Butte County was impacted by this increasing wildfire danger in 2018 by the occurrence of the Camp Fire, which was the deadliest and most destructive wildfire in California history. Vegetation landcover is an important consideration as one of the risk factors an area has for the potential occurrence of wildfires. The objective of this study is to quantify the changing wildfire risks of Butte County census tracts due to changes in vegetation cover.

Data and Methods

Data:

- National Land Cover Database: (U.S. Geological Survey) 2011 and 2021.
- California Wildfire Incidents, 2013-2023: Provides the point locations of wildfire incidents in California between 2013 and 2023 (California Department of Forestry and Fire Protection).
- Butte County Census Tract Boundaries (U.S. Census Bureau).

Methods:

- Change detection: Used in this study to portray vegetation changes in Butte County (Figure 1).
- Overlay and buffer analysis: Buffer polygons around a 1-mile radius surrounding each wildfire incident location. Vegetation data were then extracted from the land cover data (Figure 2).
- Suitability analysis: Uses multiple weighted criteria to rank and score specified locations; used in this study to create the Wildfire Risk Index (WRI) for each census tract in Butte County. The WRI was determined by weighting the vegetation types in each census tract by the percentage of occurrence of that vegetation type within the combined wildfire incident buffer zones (Figure 3). The WRI thus provides a rough estimate of the wildfire risk within a selected area due solely to its vegetation coverage (Figure 4).

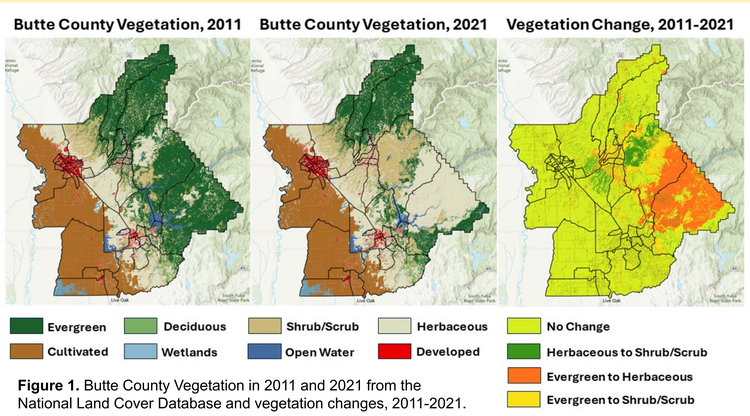

Figure 1. Butte County Vegetation in 2011 and 2021 from the National Land Cover Database and vegetation changes, 2011-2021.

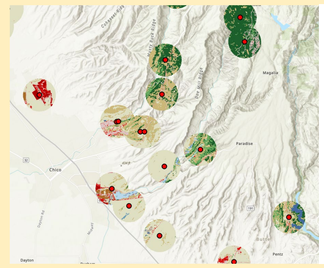

Figure 2. Example of wildfire incident locations (shown by red dots) and vegetation cover within 1-mile radius buffer zones.

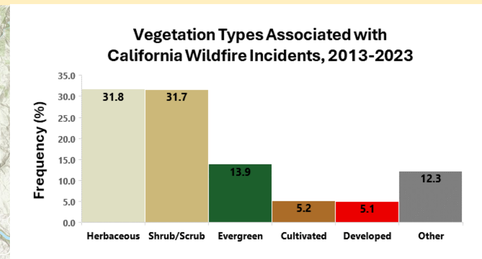

Figure 3. Frequency of occurrence of wildfire types within the combined wildfire incident buffer zones for California, 2013-2023, using the 2021 land cover data.

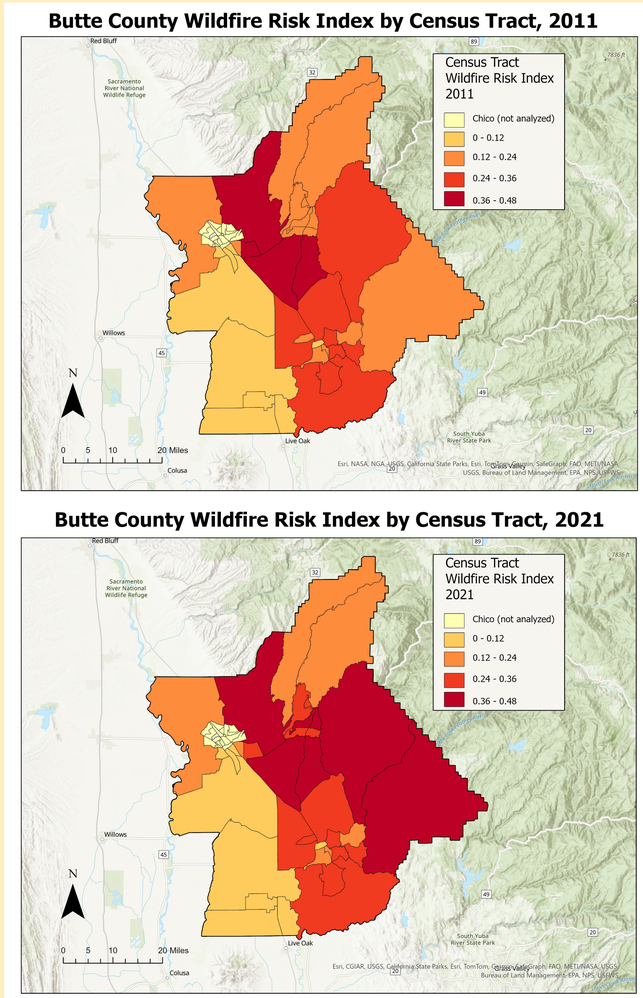

Figure 4. Wildfire Risk Index (WRI) for Butte County census tracts for 2011 and 2021. Note that evergreen-to-herbaceous and evergreen-to-shrub/scrub vegetation changes between 2011 and 2021 in eastern Butte County (Figure 1) led to higher WRI scores in these areas in 2021. Evergreen-to-shrub/scrub vegetation changes in the Paradise area due to the Camp Fire of 2018 have also increased the WRI in that area in 2021. The Chico municipal area (shown in light yellow) was not included in the analysis since very little change occurred in this developed area.

Results

- The evaluation of vegetation within the wildfire buffer zones shows that ~77% of wildfire areas coincide with just three vegetation types: herbaceous, shrub/scrub and evergreen forest (Figure 3).
- Significant vegetation changes from evergreen forest to herbaceous and from evergreen forest to scrub/shrub occurred in eastern Butte County between 2011 and 2021 (Figure 1) due to the occurrence of large wildfires in this area (Figure 5).
- These vegetation changes led to corresponding changes in the Wildfire Risk Index (Figure 4), with census tracts in eastern Butte County indicating significantly increased risk of wildfires.

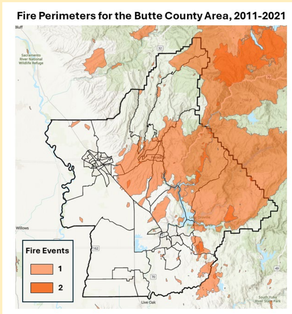

Figure 5. Fire perimeter areas for the Butte County area for the years 2011 to 2021, inclusive. Note that the wildfire burn areas in eastern Butte County and the Paradise area correspond well with the vegetation cover changes seen in Figure 1.

Discussion

- The Wildfire Risk Index (WRI) defined in this study provides meaningful information on an area's potential wildfire danger due to vegetation type alone, which is one of the major risk factors.
- The WRI has limitations in predicting where wildfires occur, since this depends upon additional factors besides vegetation, such as the prevalence of certain human activities (campfires, machinery, arson, etc.), lightning, high-wind events, topography, recent climate conditions and the recency of past fire events in an area.

Conclusions

- This study indicates that large census tracts in eastern Butte County may be more susceptible to wildfire events, due to vegetation changes that were observed between 2011 and 2021.
- Wildfire potential depends on additional factors besides vegetation, such as the prevalence of certain human activities, lightning, dry downslope wind events, topography and recent climate conditions.
- This study is a starting point of one of the major wildfire risk factors (vegetation), but a comprehensive study of wildfire potential in Butte County should include the above risk factors and more.

References

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